LESSON PLAN · GENERATIVE AI (GR 6–8) INTERMEDIATE: GEN-AI CREATORS · LEVEL 2

Lesson 5: AI for Data

Module 2 | 60 minutes | Survey · Summarise · Chart · Check | Grades 7–11 · ages 12–16

Objective: Run an anonymous class survey, ask an AI to summarise the results, make the chart yourself in a spreadsheet and check every number.

Success criteria

- I can make a fair, anonymous survey.
- I can count results in a spreadsheet.
- I can ask an AI to summarise counts and check every number.
- I can choose and label the right chart.

Key vocabulary

- **survey**: Asking many people the same question to collect data.
- **anonymous**: Without names, so nobody can be identified.
- **COUNTIF**: A spreadsheet formula that counts cells matching a rule.
- **percentage**: A part out of 100: count ÷ total × 100.
- **bar chart**: A chart of bars that compares groups.
- **finding**: A true statement the data shows.

You will need

- Projector or screen share with Kitabash and the AI tools for this lesson
- Devices: 1 per learner (laptop, tablet or phone)
- Learner accounts only where allowed: see the course safety notes
- Printed worksheet (optional)
- Project files: l05-survey-report-kit.md, l05-teacher-demo-script.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game**	Human bar chart: learners vote for a snack, count the groups and race to work out a percentage.
3 min	**Live poll**	Run the anonymous favourite-sport poll. This is the class data for today.
5 min	**Watch**	Play the lesson video; pause on the caught mistakes and ask learners to find them before revealing.
8 min	**Live demo**	Teacher enters the poll data, uses COUNTIF and SUM, asks the AI for a summary, checks every number with the class and makes a bar chart.

7 min	**Prompt Lab (pairs)**	Pairs write the summarise-my-survey prompt and improve it until every item is ticked.
5 min	**Quick competition**	Which chart fits best sort and Check the maths: first three all-correct win.
13 min	**Breakout: build the report**	In breakout rooms, learners build their sheet, get an AI summary, check it, make the chart and draft three findings. Survey to report order game for early finishers.
5 min	**Show-and-tell**	Rooms share one chart and one AI mistake they caught. Thumbs up if your AI made a maths slip.
9 min	**Probe + exit ticket**	New situation: the sleep survey, Privacy check: the class list, then the exit ticket.

Unplugged warm-up

Human bar chart (5 min): learners “stand” (or hold up a coloured card on camera) for their favourite of four snacks: samosa, idli, poha, vada pav. Count each group out loud and draw the bars on the whiteboard. Ask: what percentage chose samosa? Who can work it out first?

Teaching the idea

Data lessons show AI at its most useful and most risky. Learners collect real, anonymous data, count it in a spreadsheet (COUNTIF, SUM), then ask AI to summarise ONLY the counts — never names. AI summaries often contain arithmetic slips and overclaims (“more than half” for 48%), so every number is checked by hand: count ÷ total × 100, and percentages should add to about 100. Learners make the chart themselves in the sheet so they understand it, choosing bar (compare), pie (parts of a whole) or line (over time). Reports carry an honest label: “AI helped summarise; numbers checked by me.” Spreadsheets need a school or family account; otherwise follow the teacher’s screen.

Analogy: AI is like a fast but careless scorer at a cricket match: great at reading out the scorecard quickly, but you still check the runs add up before you announce the result.

Live demo: Before class, set up an anonymous poll (your video-call poll tool or a show of hands) with “Favourite sport: Cricket / Football / Badminton / Kabaddi”. 1) Enter the results into a sheet on the shared screen with codes S01, S02... 2) Type =COUNTIF(B2:B26, "Cricket") and fill down; add =SUM. 3) Paste ONLY the counts into the Kitabash AI helper with “Give the percentage for each, show your working, and 3 findings.” 4) Check each percentage with the class using a calculator. If the AI slips (a wrong percentage, “more than half” for under 50%), celebrate the catch. If it is all correct, ask: “Is every finding actually true?” and check the words too. 5) Insert a bar chart, add a title and axis labels.

Questions to ask

Ask	Listen for

What should you paste into an AI from a survey?	Only the counts — never names or personal details.
How do you check a percentage?	Count ÷ total × 100, and check they add to about 100.
When do you use a pie chart?	To show parts of one whole.
Is 48% “more than half”?	No — half is 50%.
Why make the chart yourself instead of copying an AI picture?	You understand it and it comes straight from checked numbers.

Differentiation

Support
Provide a ready sheet with the counts filled in; learners only add the percentage column and the chart. Calculators allowed.

Stretch
Survey two groups (for example two classes or two families) and compare them in a grouped bar chart. Ask the AI for findings and decide which ones the data truly supports.

Common misconceptions

Misconception	How to fix it
AI is a calculator, so its maths is always right	Chatbots predict text and can slip with numbers. Check every calculation.
Names make the data more accurate	Counts are all you need. Names add privacy risk and no accuracy.
Any chart will do	Bars compare, pies show parts of a whole, lines show change over time.
The most popular choice is always the majority	The top choice can be under 50%. Majority means more than half.

Assessment

Use the Prompt Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
An AI says 7 out of 20 is 40%. Is it right? Show your working.

Homework
Survey your family (or neighbours, with permission) with one anonymous question. Make a chart and write three findings. Label any AI help.

Answer key — Kitabash activities

1. Prompt Lab: summarise my survey (perform · ai lab)

Checklist: Give the counts · Give the total · Ask for percentages · Ask it to show working · Ask for findings · Ask about the chart

Strong example:

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Here are the results of our anonymous class survey on favourite sport: Cricket 12,
Football 6, Badminton 4, Kabaddi 3 (25 students). Give the percentage for each sport and
show your working. Then write 3 short findings, and tell me which chart type would show
this best and why.
```

2. Which chart fits best? (perform · categorize)

Bar chart: Comparing favourite sports in the class, Number of books read by each house team
Pie chart: How our ₹500 Diwali budget was split into parts, What share of the class walks, cycles or takes the bus
Line chart: Rainfall in each month of the monsoon, A plant's height measured every week

3. Check the maths (perform · fill blank)

Answers: 20 · 25 · wrong

4. Survey to report (perform · ordering)

Order: Write one clear, anonymous question → Collect the answers → Count them in a spreadsheet → Ask AI to summarise the counts → Check every number yourself → Make a labelled chart and write the report

5. New situation: the sleep survey (probe · multiple choice)

Answer: Work it out: 17 ÷ 30 is about 57%, so correct the report

6. Privacy check: the class list (probe · free response)

Look for: name, anonymous, count, privacy. **Example:** They should remove all names and only paste the counts, like Cricket 12, Football 6. Names are personal information, and the AI does not need them to work out percentages. Keeping the survey anonymous protects everyone's privacy.

Answer key — worksheet

1. **Key words:** survey = Asking many people the same question to collect data. · anonymous = Without names, so nobody can be identified. · COUNTIF = A spreadsheet formula that counts cells matching a rule. · percentage = A part out of 100: count ÷ total × 100.

2. **Prompt Lab: summarise my survey:** Include: Give the counts; Give the total; Ask for percentages; Ask it to show working; Ask for findings; Ask about the chart. Example: Here are the results of our anonymous class survey on favourite sport: Cricket 12, Football 6, Badminton 4, Kabaddi 3 (25 students). Give the percentage for each sport and show your working. Then write 3 short findings, and tell me which chart type would show this best and why.
3. **Which chart fits best?:** Bar chart: Comparing favourite sports in the class, Number of books read by each house team · Pie chart: How our ₹500 Diwali budget was split into parts, What share of the class walks, cycles or takes the bus · Line chart: Rainfall in each month of the monsoon, A plant's height measured every week
4. **Check the maths:** 20 · 25 · wrong
5. **Survey to report:** 1 Write one clear, anonymous question · 2 Collect the answers · 3 Count them in a spreadsheet · 4 Ask AI to summarise the counts · 5 Check every number yourself · 6 Make a labelled chart and write the report

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